



# Jeffrey Paul O.

## Applied AI Researcher | MD

Applied AI Researcher specializing in pre-training and large-scale optimization of foundational AI models. Lead Scientist behind SabiYarn (ACL 2025), with technical expertise in distributed training (ZeRO-2/3) and scaling 20B+ parameter models. Expert in framing high-dimensional healthcare and multilingual challenges into actionable, scalable R&D roadmaps. Combines rigorous benchmarking and PEFT methods with a commitment to building efficient, high-impact AI that serves the public interest.

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### Assets

- AI Research & Development
- Distributed ML Engineering & Applied AI
- AI Evaluation & Assessment
- ML System Design
- MLOps

### Professional Affiliations

- Masakhane
- Datascience Nigeria
- Igbo AI

### Grants

- ML Collective Compute Grant  
Awarded for SabiYarn-125M development.

### Social networks

- pauljeffrey
- jeffreypaul
- jeffreypaul

### Work experience

- Freelance (Contract)** Remote  
ML Researcher Since Sept. 2019
  - Distributed LLM Pre-training (Bittensor)**
    - Executed continuous pre-training of GPT-NeoX-20B and GPT-j-6B models on the bittensor dataset using **DeepSpeed ZeRO-3** across **multi-node GPU** clusters.
    - Achieved a final cross-entropy loss of 1.89 (6.62 PPL), optimizing model utility for the Bittensor network and boosted TAO token earnings by 15%.
  - AfriCOMET-Guided RLHF for Low-Resource West African Machine Translation (ongoing)**
    - Partnered with research scientists to implement an RLHF pipeline for Gemma-3-270M; optimizing performance for low-resource translation tasks
  - PEPLER + Residual Prompt Reparameterization on LLMs**
    - Collaborated on the implementation and training of prompt embeddings for explainable product recommendations. **BLEU-4** - 0.8197
  - Multi-Modal Clinical Depression Detection Model (DAIC-WOZ dataset)**
    - Built and trained a multi-modal pipeline for clinical depression diagnosis. Accuracy: 82%, Precision: 0.69, Recall: 0.7, F1-score: 0.63.
  - SignSynth: AI-Powered Speech-to-Sign Language Translator**
    - Built a multi-modal pipeline using **Transformers** for pose generation and **U-Nets** for sign language image translation (SSIM: 0.9478, MAE: 0.04)
- AISHA** Since Apr. 2026  
AI Health Advisor / Clinical Governance
  - Optimized multilingual LLM performance, boosting output quality by **20%** and reducing prompt length by **83%** via iterative ablation and clinical evaluation.
  - Engineered an agentic data pipeline for real-time conversation analysis, enabling automated health insights and dynamic plot generation, reducing administrative burden for care coordinators.
  - Developed an "LLM-as-a-judge" safety framework to rigorously validate triage accuracy, empathy, and cultural alignment in multilingual clinical interactions.
  - Designed a clinical data governance framework within a multi-tenant SaaS environment, translating research safety requirements into scalable, validated triage protocols that respect regional linguistic nuances.
- Above Health Ltd** From Oct. 2025 to Feb. 2026  
Senior AI Engineer
  - Designed and deployed a clinical AI system integrated with EHR via FHIR, enabling real-time, context-aware appointment orchestration that directly improved clinical workflows.
  - Implemented rigorous automated evaluation frameworks to mitigate hallucinations, prioritizing clinical safety and reliability in a live, member-facing environment.
- Aletheia** From Nov. 2023 to Oct. 2024  
Chief AI Scientist-SabiYarn-125M LM
  - Led end-to-end development of SabiYarn, Nigeria's first foundational decoder-only LM (8 languages), achieving superior performance over **GPT-4 (+48%)** and **LLaMA2-13B (+259%)** on regional benchmarks.
  - Architected multi-task training and data pipelines using PyTorch FSDP to optimize convergence stability and efficiency for low-resource multilingual data.
  - Published peer-reviewed research at AfricaNLP ACL 2025**, advancing AI reliability and understanding in multilingual contexts.

## Interests

- AI Research
- Explainable AI
- Health AI
- Biomedical Engineering
- Robotics
- AI in Finance

## Languages

- English

## Work experience

- Freelance

AI Engineer

Remote

From Sept. 2024 to Nov. 2025

  - RAG-Powered Customer Support Chatbot**
    - Designed and deployed a high-throughput RAG architecture (Gpt4All), translating research-grade RAG concepts into a stable production system.
    - Improved response accuracy by **30%** and successfully diverted **10%** of support volume through automated real-time resolution.
  - Brazilian BERT-Based Appliance Repair Assistant**
    - Collaborated in the development of a scalable, production-grade inference service by deploying a fine-tuned BERT model via a high-availability REST API.
    - Optimized semantic search, model latency and reliability, resulting in a direct improvement to first-time fix rates and a measurable reduction in human support overhead.
  - AI-Driven Traffic and Accident Event Analysis Engine**
    - Engineered a vision-language pipeline integrating VideoViT and multi-model LLM orchestration to classify complex accident events.
    - Achieved a **20%** improvement in classification accuracy, demonstrating expertise in experimental iteration and model optimization.
- Sienna Analytics Consulting

AI Engineer, LLM & Generative AI

From Oct. 2023 to Feb. 2024

  - Collaborated in the architectural design, development and deployment of advanced AI chatbot systems seamlessly integrating them into the business applications for 4 clients.
  - Employed strategic prompt engineering approaches improving AI responses (correctness and hallucinations) by 25%.

## Education

- College of Medicine, University Of Lagos,

MBBS

Nigeria

From Nov. 2011 to July 2018

## Tech Stack/Skills

- AI
  - Programming:** Python, Javascript..
  - Libraries & Frameworks:** Pytorch, Tensorflow, PEFT, Pydantic AI, Langchain, Langgraph, Deepspeed, huggingface transformers, accelerate, Neo4j
  - MLOps:** MLflow, Docker, Git, AWS, FastAPI.
  - Big Data & Engineering:** Pyspark.
  - Database:** Redis, MongoDB, PostgreSQL, SQL.
  - Observability & Evaluation:** LangSmith, Logfire
  - Alignment & Safety:** RLHF, Explainable AI, adversarial robustness.
  - Health AI:** Clinical Governance, Evals, Multilingual Clinical Interactions

## Certifications

- Deep Learning Specialization - Andrew Ng (2022)
- Google TensorFlow Developer (2022)

## Research Publications

- SabiYarn-125M: Advancing Low Resource Languages with Multitask NLP Pretraining**
  - [Otoibhi, J., Damilola, O., & David, O. \(2025\). SabiYarn: Advancing Low Resource Languages with Multitask NLP Pretraining. Proceedings of the Sixth Workshop on AfricaNLP 2025 \(pp. 95-107\).](#) Association for Computational Linguistics.

## Research contributions

- Automatic Post-Edit (APE) Translator with Real-time Adaptation**
  - Researched and designed a language translator with real-time learning adaptation, achieving a 0.43 BLEU-4, 0.66 TER score for domain-specific text.
- Adapting Large Language Models for Collaborative Semantic Recommendations**
  - Modified original implementation to support unique user index generation.
- Evaluating Bias in Large Language Models For African Languages**
  - Language bias in LLMs, comparing metrics between African, English and European texts.